

● MEDIUM

● EXPRESSION OF IDEAS

---

# Expression of Ideas

# 16

10 questions · ~15 min



While researching a topic, a student has taken the following notes:

- *Here I Have Returned* is a sculpture by Egyptian American artist Sherin Guirguis.
- It is a large, curved strip of wood inspired by the shape of a sistrum.
- A sistrum is a curved musical instrument played by ancient Egyptian priestesses in ceremonies.
- Guirguis says that the sculpture symbolizes “women who have lifted and supported Egyptian society and culture.”
- Overall, Guirguis wants her works to “engage audiences in a dialogue about power, agency, and social transformation.”

The student wants to use a quotation from Guirguis to explain what the sculpture represents. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Guirguis, whose works include a sculpture that is a large, curved strip of wood, has explained that she wants her work to create a dialogue with audiences.
- B. Inspired by the sistrum played by Egyptian priestesses, *Here I Have Returned* symbolizes “women who have lifted and supported Egyptian society and culture,” according to Guirguis.
- C. According to Guirguis, the curved strip of wood used in *Here I Have Returned* was inspired by the sistrum, a musical instrument played by ancient Egyptian priestesses in ceremonies.
- D. Guirguis, the sculptor of *Here I Have Returned*, wants her works to “engage audiences in a dialogue about power, agency, and social transformation.”

While researching a topic, a student has taken the following notes:

- Meteorites found on Earth are divided into two categories.
- A meteorite that was observed falling to Earth before being recovered is known as a meteorite fall.
- All other meteorites found on Earth are known as meteorite finds.
- There have been about 1,200 recorded meteorite falls.
- There have been over 60,000 recorded meteorite finds.

The student wants to contrast the number of meteorite falls with the number of meteorite finds. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. A meteorite that was observed falling to Earth before being recovered is known as a meteorite fall; all others are known as meteorite finds.
- B. Meteorites found on Earth are divided into two categories: meteorite falls and meteorite finds.
- C. There have been about 1,200 recorded meteorite falls, or meteorites observed falling to Earth.
- D. While there have been only about 1,200 recorded meteorite falls, there have been over 60,000 meteorite finds.

While researching a topic, a student has taken the following notes:

- 1926: The US Congress gave the US Commerce Department authority to regulate safety standards in the fledgling commercial airline industry.
- 1938: Congress transferred this authority to a new independent government agency called the Civil Aeronautics Authority (CAA).
- 1958: Congress transferred authority from the CAA to the newly established Federal Aviation Administration (FAA).
- The FAA's first administrator, Elwood R. Quesada, updated safety standards and technologies for the era of commercial jets.
- The FAA remains the regulatory authority for airline safety.

The student wants to specify the order in which different government entities were given the authority to regulate airline safety in the US. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The CAA had the authority to regulate safety for US airlines from 1938 until 1958, at which point authority was transferred to the US Commerce Department by Elwood R. Quesada.
- B. The FAA, CAA, and the US Commerce Department all had the authority to regulate US airline safety, but they possessed this authority at different times.
- C. The FAA has regulated airline safety since it was established by the US Congress in 1958.
- D. The authority to regulate US airline safety transferred from the US Commerce Department to the CAA in 1938, then from the CAA to the FAA in 1958.

While researching a topic, a student has taken the following notes:

- Chromosomes are cellular structures that contain genes.
- Genes carry critical instructions for determining an organism's physical traits.
- Members of the same species typically have the same number of chromosomes.
- The pineapple (*Ananas comosus*) and the melon (*Cucumis melo*) are species of fruits.
- The pineapple has fifty chromosomes.
- The melon has twenty-four chromosomes.

The student wants to specify how many chromosomes the pineapple has. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The pineapple's chromosomes contain genes, which are critical to determining an organism's physical traits.
- B. The pineapple (*Ananas comosus*) has fifty chromosomes.
- C. The pineapple (*Ananas comosus*) and the melon (*Cucumis melo*) both have chromosomes, but the pineapple has more than the melon does.
- D. The melon, a species of fruit, has twenty-four structures called chromosomes.

While researching a topic, a student has taken the following notes:

- When medical students mention their patients on social media, they may violate patient confidentiality.
- Terry Kind led a study to determine how many medical schools have student policies that mention social media use.
- Kind and her team reviewed 132 medical school websites, examining publicly available student policies.
- Only thirteen medical schools had guidelines that explicitly mention social media, and only five defined what constitutes acceptable social media use.

The student wants to emphasize the study's methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The student policies of 132 medical schools can be found online, according to research by Terry Kind.
- B. To find out how many medical schools have guidelines about student social media use, Terry Kind and her team examined the student policies of 132 medical schools.
- C. Out of 132 medical schools, only thirteen had student policies that mentioned social media, and only five specified what use was acceptable.
- D. Terry Kind and her team wanted to know how many medical schools have student social media policies in place about protecting patient confidentiality.

While researching a topic, a student has taken the following notes:

- Traditionally, manufacturers have dyed denim jeans blue by dipping them in a solution containing indigo powder.
- Indigo doesn't dissolve in just water, so manufacturers must mix hazardous chemicals with water to dissolve the powder.
- Textile researcher Smriti Rai discovered a process for dyeing blue jeans without chemicals.
- Rai added indigo powder to a hydrogel containing nanocellulose and produced a dye that could be spread directly onto the denim.
- Nanocellulose is a natural, plant-based substance that separates the molecules of indigo powder.

The student wants to emphasize a difference between the two approaches to dyeing blue jeans. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Though created using a different process, Rai's dye contains the same ingredient as the dye produced by blue jean manufacturers.
- B. Nanocellulose is a natural, plant-based substance that separates the molecules of indigo powder, which doesn't dissolve in water.
- C. The traditional approach to dyeing blue jeans is to dip them in a solution containing hazardous chemicals.
- D. Rai's approach substitutes a natural, plant-based substance for the hazardous chemicals that manufacturers have traditionally used.



While researching a topic, a student has taken the following notes:

- Bharati Mukherjee was an Indian-born author of novels and short stories.
- She published the novel *The Holder of the World* in 1993.
- A central character in the novel is a woman living in twentieth-century United States.
- Another central character is a woman living in seventeenth-century India.

The student wants to introduce the novel *The Holder of the World* to an audience already familiar with Bharati Mukherjee. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Bharati Mukherjee's settings include both twentieth-century United States and seventeenth-century India.
- B. In addition to her novel *The Holder of the World*, which was published in 1993, Indian-born author Bharati Mukherjee wrote other novels and short stories.
- C. Bharati Mukherjee's novel *The Holder of the World* centers around two women, one living in twentieth-century United States and the other in seventeenth-century India.
- D. *The Holder of the World* was not the only novel written by Indian-born author Bharati Mukherjee.

While researching a topic, a student has taken the following notes:

- A lever is a simple machine consisting of a rigid beam and a fulcrum.
- The fulcrum is the point about which the beam pivots.
- The input force (effort) is the force applied to the lever.
- The output force (load) is the force that the lever exerts on another object.
- In first-class levers, the fulcrum is located between the effort and the load.
- In second-class levers, the load is located between the effort and the fulcrum.

The student wants to contrast first-class levers and second-class levers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. In levers, the effort is the force applied to the lever; the load, in contrast, is the force that the lever exerts on another object.
- B. In first-class and second-class levers, the fulcrum and the load are in different locations.
- C. First-class levers are simple machines consisting of a rigid beam and a fulcrum, but then again, the same is true of second-class levers.
- D. In first-class levers, the fulcrum is located between the effort and the load, but in second-class levers, the load is located between the effort and the fulcrum.

While researching a topic, a student has taken the following notes:

- Species belonging to the Orchidaceae (orchid) family can be found in both tropical and temperate environments.
- Orchidaceae species diversity has not been well studied in temperate forests, such as those in Oaxaca, Mexico.
- Arelee Estefanía Muñoz-Hernández led a study to determine how many different Orchidaceae species are present in the forests of Oaxaca.
- Muñoz-Hernández and her team collected orchids each month for a year at a site in Oaxaca.
- Seventy-four Orchidaceae species were present at the site.

The student wants to present the study and its findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. A study led by Arelee Estefanía Muñoz-Hernández identified a total of 74 Orchidaceae species in the temperate forests of Oaxaca, Mexico.
- B. There are orchids in many environments, but there are 74 Orchidaceae species in Oaxaca, Mexico.
- C. Oaxaca, Mexico, is home to temperate forests containing 74 Orchidaceae species.
- D. Arelee Estefanía Muñoz-Hernández and her team wanted to know how many different Orchidaceae species are present in the forests of Oaxaca, Mexico, so they conducted a study to collect orchids.

While researching a topic, a student has taken the following notes:

- Physicist Muluneh Abebe was working on a garment suited for both warm and cold conditions.
- He analyzed the emissivity, or ability to emit heat, of the materials he planned to use.
- Abebe found that reflective metal fibers emitted almost no heat and had an emissivity of 0.02.
- He found that silicon carbide fibers absorbed large amounts of heat and had an emissivity of 0.74.
- The amount of heat a material absorbs is equal to the amount of heat it emits.

The student wants to contrast the emissivity of reflective metal fibers with that of silicon carbide fibers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The ability of reflective metal fibers and silicon carbide fibers to emit heat was determined by an analysis of each material's emissivity.
- B. The amount of heat a material absorbs is equal to the amount it emits, as evidenced in Abebe's analyses.
- C. Though the reflective metal fibers and silicon carbide fibers had different rates of emissivity, Abebe planned to use both in a garment.
- D. Whereas the reflective metal fibers had an emissivity of just 0.02, the silicon carbide fibers absorbed large amounts of heat, resulting in an emissivity of 0.74.